

Vibration of continuous systems

Continuous systems will have **infinite** degrees of freedom and **infinite number of frequencies**.

Longitudinal (Axial) vibration of bars

Consider an elastic bar of length l with varying cross-sectional area $A(x)$ and density ρ as shown. The forces acting on the small element of the bar are given by P and $P + dP$ with

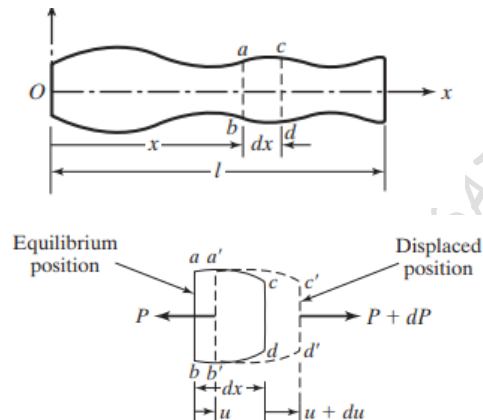
$$P = \sigma A = EA(x) \frac{\partial u}{\partial x}$$

Applying the equation of motion to the element

$$(\rho A(x) dx) \ddot{u} = P + dP - P = \frac{\partial P}{\partial x} dx = \frac{\partial}{\partial x} \left(EA(x) \frac{\partial u}{\partial x} \right) dx$$

For a constant axial rigidity EA , $\rho A \frac{\partial^2 u}{\partial t^2} = EA \frac{\partial^2 u}{\partial x^2}$

$$c^2 \frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2} \quad \ll \gg \quad c = \sqrt{\frac{E}{\rho}}$$



The governing equation is a second order partial differential equation. The displacement $u(x, t)$ is a function of x and t . Let us assume that the solution can be written as two functions of x and t (variable separable form)

$$u(x, t) = X(x) T(t)$$

Where X and T are functions of x alone and t alone respectively. Then,

$$\frac{\partial^2 u}{\partial x^2} = T \frac{d^2 X}{dx^2} \quad \text{and} \quad \frac{\partial^2 u}{\partial t^2} = X \frac{d^2 T}{dt^2}$$

Therefore the differential equation becomes

$$c^2 T \frac{d^2 X}{dx^2} = X \frac{d^2 T}{dt^2} \quad \gg \gg \gg \quad c^2 \frac{1}{X} \frac{d^2 X}{dx^2} = \frac{1}{T} \frac{d^2 T}{dt^2}$$

The left hand side is a function of x alone and the right hand side is a function of t alone. Therefore each function should be a constant.

$$c^2 \frac{1}{X} \frac{d^2 X}{dx^2} = \frac{1}{T} \frac{d^2 T}{dt^2} = -\omega^2$$

$$\frac{1}{T} \frac{d^2 T}{dt^2} = -\omega^2 \quad \gg \gg \quad \frac{d^2 T}{dt^2} + \omega^2 T = 0$$

Assuming a solution $T(t) = Ae^{pt}$

The constant is assumed to be negative. If it is positive, the roots of the characteristic equation will be $\pm \omega$, which means the response will grow exponentially with time which is physically not possible.

$$Ap^2 e^{pt} + \omega^2 Ae^{pt} = 0 \quad \gg \gg \gg \quad Ae^{pt}(p^2 + \omega^2) = 0 \quad \gg \gg \gg \quad p^2 + \omega^2 = 0 \quad \gg \gg \gg \quad p = \pm i\omega$$

$$T(t) = A_1 e^{i\omega t} + A_2 e^{-i\omega t}$$

The solution of this equation is $T(t) = A \sin \omega t + B \cos \omega t$. The arbitrary constants A and B are to be found out from initial conditions (u , and $\frac{\partial u}{\partial t}$ at $t = 0$). Similarly,

$$c^2 \frac{1}{X} \frac{d^2 X}{dx^2} = -\omega^2 \quad \gg \gg \gg \quad \frac{d^2 X}{dx^2} + \left(\frac{\omega}{c}\right)^2 X = 0$$

$$X(x) = C_1 \sin\left(\frac{\omega}{c}x\right) + C_2 \cos\left(\frac{\omega}{c}x\right)$$

The arbitrary constants C_1 and C_2 are to be evaluated from the boundary conditions u (displacement) and $\frac{\partial u}{\partial x}$ (strain) at the boundaries

The complete solution is $u(x, t) = \left\{ C_1 \sin\left(\frac{\omega}{c}x\right) + C_2 \cos\left(\frac{\omega}{c}x\right) \right\} \{ A \sin \omega t + B \cos \omega t \}$

The first term on the right side is showing the mode shape and the second term shows the variation of the mode shape with respect to time.

#Example 1

Determine the natural frequencies of axial free vibration of a cantilever bar with modulus of elasticity E , and density ρ , and uniform cross sectional area A .

Let us consider the equation of the mode shape

$$u = C_1 \sin\left(\frac{\omega}{c}\right)x + C_2 \cos\left(\frac{\omega}{c}\right)x$$

$$\frac{\partial u}{\partial x} = \left(\frac{\omega}{c}\right) \left\{ C_1 \cos\left(\frac{\omega}{c}\right)x - C_2 \sin\left(\frac{\omega}{c}\right)x \right\}$$

The boundary conditions (u , and $\frac{\partial u}{\partial x}$) of a cantilever bar are



Displacement at the fixed end, $u_{(0,t)} = 0$

Strain at the free end, $\frac{\partial u}{\partial x}(l,t) = 0$

$$0 = 0 + C_2 \gg \gg C_2 = 0$$

$$0 = \left(\frac{\omega}{c}\right) \left\{ C_1 \cos\left(\frac{\omega}{c}\right)l \right\} \gg \gg \cos\left(\frac{\omega}{c}\right)l = 0 \gg \gg \left(\frac{\omega}{c}\right)l = \left(\frac{2n+1}{2}\right)\pi$$

$$\omega = \left(\frac{2n+1}{2}\right)\left(\frac{\pi c}{l}\right), n = 0, 1, 2, \dots$$

The general solution is given by the superposition of all such solutions.

$$u(x,t) = \sum_{n=0}^{\infty} C_{1n} \sin\left(\frac{\omega_n}{c}\right)x \{ A_n \sin \omega_n t + B_n \cos \omega_n t \} = \sum_{n=0}^{\infty} \sin\left(\frac{2n+1}{2}\right)\left(\frac{\pi x}{l}\right) \left\{ C_n \sin\left(\frac{2n+1}{2}\right)\left(\frac{\pi c t}{l}\right) + D_n \cos\left(\frac{2n+1}{2}\right)\left(\frac{\pi c t}{l}\right) \right\}$$

$$\dot{u}(x,t) = \sum_{n=0}^{\infty} \sin\left(\frac{2n+1}{2}\right)\left(\frac{\pi x}{l}\right) \left\{ C_n \left(\frac{2n+1}{2}\right)\left(\frac{\pi c}{l}\right) \cos\left(\frac{2n+1}{2}\right)\left(\frac{\pi c t}{l}\right) - D_n \left(\frac{2n+1}{2}\right)\left(\frac{\pi c}{l}\right) \sin\left(\frac{2n+1}{2}\right)\left(\frac{\pi c t}{l}\right) \right\}$$

Now substituting the initial conditions, $u(x,0) = u_0(x)$, and $\frac{\partial u}{\partial t}(x,0) = \dot{u}_0(x)$

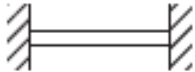
$$u_0(x) = \sum_{n=0}^{\infty} D_n \sin\left(\frac{2n+1}{2}\right)\left(\frac{\pi x}{l}\right) \ll \gg \dot{u}_0(x) = \sum_{n=0}^{\infty} C_n \left(\frac{2n+1}{2}\right)\left(\frac{\pi c}{l}\right) \sin\left(\frac{2n+1}{2}\right)\left(\frac{\pi x}{l}\right)$$

These are Fourier sine series in the interval $0 \leq x \leq l$. The values of C_n and D_n are obtained by multiplying the above equations by $\sin\left(\frac{2n+1}{2}\right)\left(\frac{\pi x}{l}\right)$ and integrating with respect to x in the interval 0 to l .

$$D_n = \frac{2}{l} \int_0^l u_0(x) \sin\left(\frac{2n+1}{2}\right)\left(\frac{\pi x}{l}\right) dx \ll \gg C_n = \frac{2}{\left(\frac{2n+1}{2}\right)\pi c} \int_0^l \dot{u}_0(x) \sin\left(\frac{2n+1}{2}\right)\left(\frac{\pi x}{l}\right) dx$$

For Torsional system, the wave equation is

$$c^2 \frac{\partial^2 \theta}{\partial x^2} = \frac{\partial^2 \theta}{\partial t^2} \gg \gg \text{where } c = \sqrt{\frac{G}{\rho}}$$

End conditions of bar	Boundary Conditions	Frequency equation	Natural frequency
 <i>Fixed Free</i>	$u(0, t) = 0$ $\frac{\partial u}{\partial x}(l, t) = 0$	$\cos\left(\frac{\omega l}{c}\right) = 0$	$\omega_n = \left(\frac{2n+1}{2}\right) \frac{\pi c}{l}$ $n = 0, 1, 2 \dots$
 <i>Free Free</i>	$\frac{\partial u}{\partial x}(0, t) = 0$ $\frac{\partial u}{\partial x}(l, t) = 0$	$\sin\left(\frac{\omega l}{c}\right) = 0$	$\omega_n = \frac{n\pi c}{l}$ $n = 0, 1, 2 \dots$
 <i>Fixed Fixed</i>	$u(0, t) = 0$ $u(l, t) = 0$	$\sin\left(\frac{\omega l}{c}\right) = 0$	$\omega_n = \frac{n\pi c}{l}$ $n = 0, 1, 2 \dots$

Exercise 1

Determine the natural frequencies of axial free vibration of a fixed-fixed bar with modulus of elasticity E , and density ρ , and uniform cross sectional area A .

Exercise 2

Determine the natural frequencies of axial free vibration of a free-free bar with modulus of elasticity E , and density ρ , and uniform cross sectional area A .

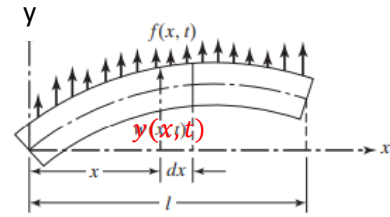
Lateral Vibration of Beams

Consider a beam vibrating in the transverse direction. If an infinitesimally small element is considered, the equation of motion in the z-direction can be written as

$$(\rho A dx) \ddot{y} = \Sigma F_z = -(V + dV) + V + f dx$$

$$(\rho A dx) \ddot{y} = -dV + f dx = -\frac{\partial V}{\partial x} dx + f dx$$

$$\Sigma M_y = 0 \gg M + dM - M - V dx - f \frac{dx^2}{2} = 0 \gg dM = V dx$$

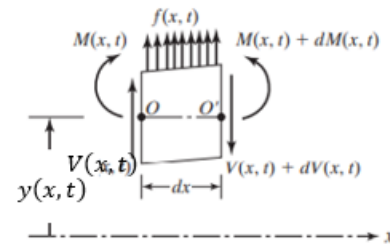


$$\frac{\partial M}{\partial x} dx = V dx \gg \frac{\partial M}{\partial x} = V$$

For free vibrations, $f(x, t) = 0$

$$(\rho A dx) \frac{\partial^2 y}{\partial t^2} = -\frac{\partial V}{\partial x} dx = -\frac{\partial}{\partial x} \left(\frac{\partial M}{\partial x} \right) dx$$

$$(\rho A) \frac{\partial^2 y}{\partial t^2} = -\frac{\partial^2}{\partial x^2} (M) = -\frac{\partial^2}{\partial x^2} \left\{ EI \frac{\partial^2 y}{\partial x^2} \right\}$$



For constant flexural rigidity EI

$$\frac{\partial^2 y}{\partial t^2} = -\left(\frac{EI}{\rho A} \right) \left\{ \frac{\partial^4 y}{\partial x^4} \right\} \ll \gg c^2 \frac{\partial^4 y}{\partial x^4} + \frac{\partial^2 y}{\partial t^2} = 0 \ll \gg c = \sqrt{\frac{EI}{\rho A}}$$

The governing equation is a fourth derivative of x and second derivative in t. There should be four boundary conditions and two initial conditions. The displacement $y(x, t)$ is a function of x and t. Let us assume that the solution can be written as two functions of x and t (variable separable form)

$$y(x, t) = X(x) T(t)$$

Where X and T are functions of x alone and t alone respectively. Then,

$$\frac{\partial^4 y}{\partial x^4} = T \frac{d^4 X}{dx^4} \quad \text{and} \quad \frac{\partial^2 y}{\partial t^2} = X \frac{d^2 T}{dt^2}$$

Therefore the differential equation becomes

$$X \frac{d^2 T}{dt^2} + c^2 T \frac{d^4 X}{dx^4} = 0 \gg \gg \gg \frac{1}{T} \frac{d^2 T}{dt^2} = -c^2 \frac{1}{X} \frac{d^4 X}{dx^4}$$

The left hand side is a function of x alone and the right hand side is a function of t alone. Therefore each function should be a constant.

$$\frac{1}{T} \frac{d^2 T}{dt^2} = -\omega^2 = -c^2 \frac{1}{X} \frac{d^4 X}{dx^4}$$

$$\frac{d^2 T}{dt^2} + \omega^2 T = 0$$

Assuming a solution $T(t) = Ae^{pt}$

$$Ap^2 e^{pt} + \omega^2 e^{pt} = 0 \gg \gg Ae^{pt}(p^2 + \omega^2) = 0 \gg \gg p^2 + \omega^2 = 0 \gg \gg p = \pm i\omega$$

$$T(t) = A_1 e^{i\omega t} + A_2 e^{-i\omega t}$$

The solution of this equation is $T(t) = A \sin \omega t + B \cos \omega t$. The arbitrary constants A and B are to be found out from initial conditions (y , and $\frac{\partial y}{\partial t}$ at $t = 0$).

Similarly,

$$-c^2 \frac{1}{X} \frac{d^4 X}{dx^4} = -\omega^2 \gg \gg \frac{d^4 X}{dx^4} - \left(\frac{\omega}{c} \right)^2 X = 0$$

Assuming a solution $X(x) = Ae^{px}$

$$Ap^4 e^{px} - \left(\frac{\omega}{c}\right)^2 e^{px} = 0 \ggg Ae^{px}(p^4 - \beta^4) = 0 \ggg p^4 - \beta^4 = 0 \ggg p = \pm\beta, \pm i\beta$$

$$\beta^4 = \left(\frac{\omega}{c}\right)^2 \gg \beta^2 = \left(\frac{\omega}{c}\right) \ggg \beta = \sqrt{\frac{\omega}{c}}$$

$$X(x) = A_1 e^{\beta x} + A_2 e^{-\beta x} + A_3 e^{i\beta x} + A_4 e^{-i\beta x}$$

$$\cosh \beta x = \frac{e^{\beta x} + e^{-\beta x}}{2} \gg \sinh \beta x = \frac{e^{\beta x} - e^{-\beta x}}{2} \lll e^{\beta x} = \frac{\cosh \beta x + \sinh \beta x}{2} \lll e^{-\beta x} = \frac{\cosh \beta x - \sinh \beta x}{2}$$

$$X(x) = A_1 \left(\frac{\cosh \beta x + \sinh \beta x}{2} \right) + A_2 \left(\frac{\cosh \beta x - \sinh \beta x}{2} \right) + A_3 (\cos \beta x + i \sin \beta x) + A_4 (\cos \beta x - i \sin \beta x)$$

$$X(x) = C_1 \cosh \beta x + C_2 \sinh \beta x + C_3 \cos \beta x + C_4 \sin \beta x$$

$$X(x) = C_1 \cosh \sqrt{\frac{\omega}{c}} x + C_2 \sinh \sqrt{\frac{\omega}{c}} x + C_3 \cos \sqrt{\frac{\omega}{c}} x + C_4 \sin \sqrt{\frac{\omega}{c}} x$$

The arbitrary constants can be evaluated by specifying conditions $y, \frac{\partial y}{\partial x}, \frac{\partial^2 y}{\partial x^2}, \frac{\partial^3 y}{\partial x^3}$ at the boundaries.

$y \rightarrow \text{deflection}$	$\frac{\partial y}{\partial x} \rightarrow \text{slope}$	$\frac{\partial^2 y}{\partial x^2} \rightarrow \text{bending moment}$ $EI \frac{\partial^2 y}{\partial x^2} = M$	$\frac{\partial^3 y}{\partial x^3} \rightarrow \text{shear force}$ $V = \frac{\partial M}{\partial x} = EI \frac{\partial^3 y}{\partial x^3}$
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1. Fixed end	Displacement=0 Slope =0	$y = 0, \quad \frac{\partial y}{\partial x} = 0$
2. Free end	Bending moment=0 Shear force=0	$\frac{\partial^2 y}{\partial x^2} = 0, \quad \frac{\partial^3 y}{\partial x^3} = 0$
3. Hinged end	Displacement=0 Bending Moment=0	$y = 0, \quad \frac{\partial^2 y}{\partial x^2} = 0$

The complete solution is

$$y(x, t) = \left\{ C_1 \cosh \sqrt{\frac{\omega}{c}} x + C_2 \sinh \sqrt{\frac{\omega}{c}} x + C_3 \cos \sqrt{\frac{\omega}{c}} x + C_4 \sin \sqrt{\frac{\omega}{c}} x \right\} \{A \sin \omega t + B \cos \omega t\}$$

#Example 1

Determine the frequency equation of free transverse vibration of a cantilever beam of length l , cross sectional area A , and flecural rigidity EI



Let us consider the equation of mode shape

$$y = C_1 \cosh \beta x + C_2 \sinh \beta x + C_3 \cos \beta x + C_4 \sin \beta x$$

$$\frac{\partial y}{\partial x} = C_1 \beta \sinh \beta x + C_2 \beta \cosh \beta x - C_3 \beta \sin \beta x + C_4 \beta \cos \beta x$$

$$\frac{\partial^2 y}{\partial x^2} = C_1 \beta^2 \cosh \beta x + C_2 \beta^2 \sinh \beta x - C_3 \beta^2 \cos \beta x - C_4 \beta^2 \sin \beta x$$

$$\frac{\partial^3 y}{\partial x^3} = C_1 \beta^3 \sinh \beta x + C_2 \beta^3 \cosh \beta x + C_3 \beta^3 \sin \beta x - C_4 \beta^3 \cos \beta x$$

The boundary conditions are

At the fixed end ($x = 0$) $y = 0$, $\frac{\partial y}{\partial x} = 0$

At the free end ($x = l$) $\frac{\partial^2 y}{\partial x^2} = 0$, $\frac{\partial^3 y}{\partial x^3} = 0$

$$0 = C_1 \cosh 0 + C_2 \sinh 0 + C_3 \cos 0 + C_4 \sin 0 \gggg 0 = C_1 + C_3 \gggg C_3 = -C_1$$

$$0 = C_1 \beta \sinh 0 + C_2 \beta \cosh 0 - C_3 \beta \sin 0 + C_4 \beta \cos 0 \gggg 0 = C_2 + C_4 \gggg C_4 = -C_2$$

$$0 = C_1 \beta^2 \cosh \beta l + C_2 \beta^2 \sinh \beta l - C_3 \beta^2 \cos \beta l - C_4 \beta^2 \sin \beta l \gg C_1 (\cosh \beta l + \cos \beta l) + C_2 (\sinh \beta l + \sin \beta l) = 0$$

$$0 = C_1 \beta^3 \sinh \beta l + C_2 \beta^3 \cosh \beta l + C_3 \beta^3 \sin \beta l - C_4 \beta^3 \cos \beta l \gg C_1 (\sinh \beta l - \sin \beta l) + C_2 (\cosh \beta l + \cos \beta l) = 0$$

$$\begin{bmatrix} (\cosh \beta l + \cos \beta l) & (\sinh \beta l + \sin \beta l) \\ (\sinh \beta l - \sin \beta l) & (\cosh \beta l + \cos \beta l) \end{bmatrix} \begin{Bmatrix} C_1 \\ C_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

The condition for the system of linear homogeneous equations to have non-trivial solutions is

$$\begin{vmatrix} (\cosh \beta l + \cos \beta l) & (\sinh \beta l + \sin \beta l) \\ (\sinh \beta l - \sin \beta l) & (\cosh \beta l + \cos \beta l) \end{vmatrix} = 0$$

$$(\cosh \beta l + \cos \beta l)(\cosh \beta l + \cos \beta l) - (\sinh \beta l + \sin \beta l)(\sinh \beta l - \sin \beta l) = 0$$

$$(\cosh^2 \beta l + 2 \cosh \beta l \cos \beta l + \cos^2 \beta l) - (\sinh^2 \beta l - \sin^2 \beta l) = 0$$

$$(\cosh^2 \beta l - \sinh^2 \beta l) + (\cos^2 \beta l + \sin^2 \beta l) + 2 \cosh \beta l \cos \beta l = 0$$

$$2 \cosh \beta l \cos \beta l = -2$$

$$\cosh \beta l \cos \beta l = -1$$

$$\cosh \beta l \cos \beta l = -1$$

$$\beta_1 l = 1.875104$$

$$\beta_2 l = 4.694091$$

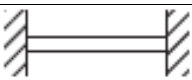
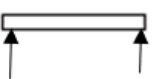
$$\beta_3 l = 7.854757$$

$$\beta_4 l = 10.995541$$

$$\begin{aligned} \beta &= \sqrt{\frac{\omega}{c}} \gg \omega = \beta^2 c = \beta^2 \sqrt{\frac{EI}{\rho A}} = \frac{(\beta l)^2}{l^2} \sqrt{\frac{EI}{\rho A}} \\ &= (\beta l)^2 \sqrt{\frac{EI}{\rho A l^4}} \end{aligned}$$

$$\omega_i = (\beta_i l)^2 \sqrt{\frac{EI}{\rho A l^4}}$$

The fundamental frequency of the beam is $\omega_1 = (1.875104)^2 \sqrt{\frac{EI}{\rho A l^4}} = 3.516015 \sqrt{\frac{EI}{\rho A l^4}}$

End conditions of beam	Boundary Conditions	Frequency equation	
 <i>Fixed Free</i>	At the fixed end $y(0, t) = 0, \frac{\partial y}{\partial x}(0, t) = 0$ At the free end $\frac{\partial^2 y}{\partial x^2}(l, t) = 0, \frac{\partial^3 y}{\partial x^3}(l, t) = 0$	$\cosh \beta_n l \cos \beta_n l = -1$	$\beta_1 l = 1.875$ $\beta_2 l = 4.694$ $\beta_3 l = 7.854$ $\beta_4 l = 10.995$
 <i>Free Free</i>	At the free ends $\frac{\partial^2 y}{\partial x^2}(0, t) = 0, \frac{\partial^3 y}{\partial x^3}(0, t) = 0$ $\frac{\partial^2 y}{\partial x^2}(l, t) = 0, \frac{\partial^3 y}{\partial x^3}(l, t) = 0$	$\cosh \beta_n l \cos \beta_n l = 1$	$\beta_1 l = 4.730$ $\beta_2 l = 7.853$ $\beta_3 l = 10.995$ $\beta_4 l = 14.137$
 <i>Fixed Fixed</i>	At the fixed ends $y(0, t) = 0, \frac{\partial y}{\partial x}(0, t) = 0$ $y(l, t) = 0, \frac{\partial y}{\partial x}(l, t) = 0$	$\cosh \beta_n l \cos \beta_n l = 1$	$\beta_1 l = 4.730$ $\beta_2 l = 7.853$ $\beta_3 l = 10.995$ $\beta_4 l = 14.137$
 <i>pinned pinned</i>	At the pinned ends $y(0, t) = 0, \frac{\partial^2 y}{\partial x^2}(0, t) = 0$ $y(l, t) = 0, \frac{\partial^2 y}{\partial x^2}(l, t) = 0$	$\sin \beta_n l = 0$	$\beta_1 l = \pi$ $\beta_2 l = 2\pi$ $\beta_3 l = 3\pi$ $\beta_4 l = 4\pi$
 <i>fixed pinned</i>	At the fixed ends $y(0, t) = 0, \frac{\partial y}{\partial x}(0, t) = 0$ At the pinned ends $y(l, t) = 0, \frac{\partial^2 y}{\partial x^2}(l, t) = 0$	$\tan \beta_n l - \tanh \beta_n l = 0$	$\beta_1 l = 3.927$ $\beta_2 l = 7.069$ $\beta_3 l = 10.210$ $\beta_4 l = 13.352$

Exercise 1

Determine the frequency equation of free transverse vibration of a simply supported beam of length l , cross sectional area A , and flexural rigidity EI .

Exercise 2

Determine the frequency equation of free transverse vibration of a fixed-fixed beam of length l , cross sectional area A , and flexural rigidity EI .

Exercise 3

Determine the frequency equation of free transverse vibration of a fixed-pinned beam of length l , cross sectional area A , and flexural rigidity EI .

Exercise 4

Determine the frequency equation of free transverse vibration of a free-free beam of length l , cross sectional area A , and flexural rigidity EI .

Reference:

Mechanical Vibrations, 6/e, Singiresu S. Rao, Pearson Global Edition